# AI & Machine Learning Curriculum

**Deep Learning & GenAI Specialization**
Updated: July 08, 2026

## Course Structure

| Module | Topic | Duration | Level |
|---|---|---|---|
| 1 | Python Fundamentals | 2 weeks | Beginner |
| 2 | NumPy & Pandas | 2 weeks | Beginner |
| 3 | Data Visualization | 1 week | Beginner |
| 4 | Statistics & Probability | 2 weeks | Intermediate |
| 5 | Machine Learning Basics | 3 weeks | Intermediate |
| 6 | Supervised Learning | 3 weeks | Intermediate |
| 7 | Unsupervised Learning | 2 weeks | Intermediate |
| 8 | Neural Networks | 4 weeks | Advanced |
| 9 | Deep Learning | 4 weeks | Advanced |
| 10 | Computer Vision | 3 weeks | Advanced |
| 11 | Natural Language Processing | 4 weeks | Advanced |
| 12 | Generative AI & LLMs | 4 weeks | Advanced |
| 13 | RAG & Prompt Engineering | 2 weeks | Advanced |
| 14 | AI Applications & Deployment | 2 weeks | Advanced |

## Learning Outcomes

• Master Python programming for data science and AI
• Understand machine learning algorithms and their applications
• Build and train deep learning models
• Work with state-of-the-art generative AI models
• Implement RAG systems and prompt engineering techniques
• Deploy AI models to production
• Handle real-world datasets and preprocessing
• Perform exploratory data analysis and visualization

## Prerequisites

• Basic understanding of mathematics (algebra, calculus)
• Familiarity with programming concepts (loops, functions, data structures)
• Basic knowledge of statistics
• Linux/Command line basics (helpful but not mandatory)

## Tools & Technologies

• Python 3.8+
• Jupyter Notebook & VS Code
• TensorFlow & PyTorch
• Scikit-learn
• Pandas & NumPy
• Matplotlib & Seaborn
• OpenAI API & Google Generative AI
• LangChain
• Docker & Git